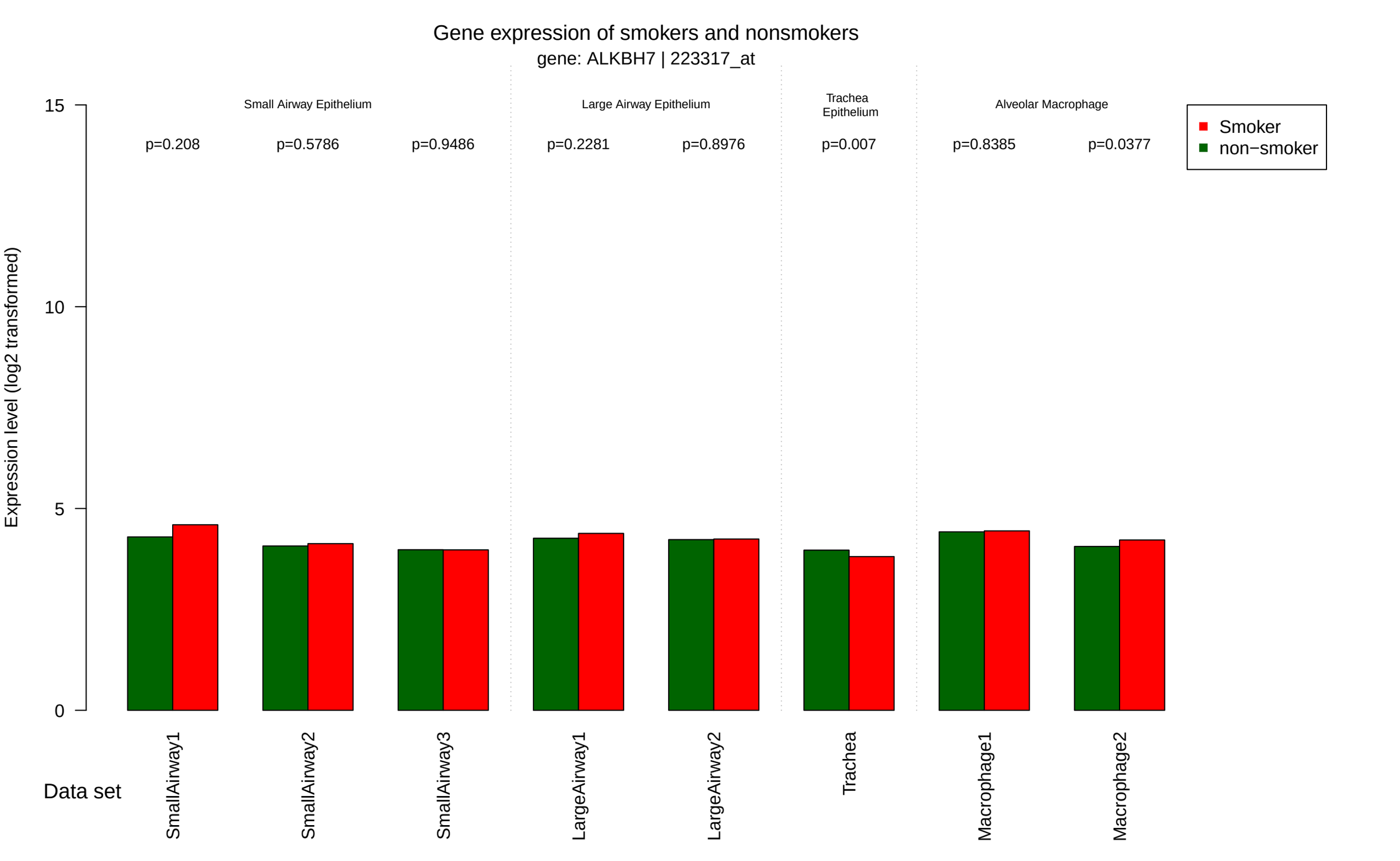
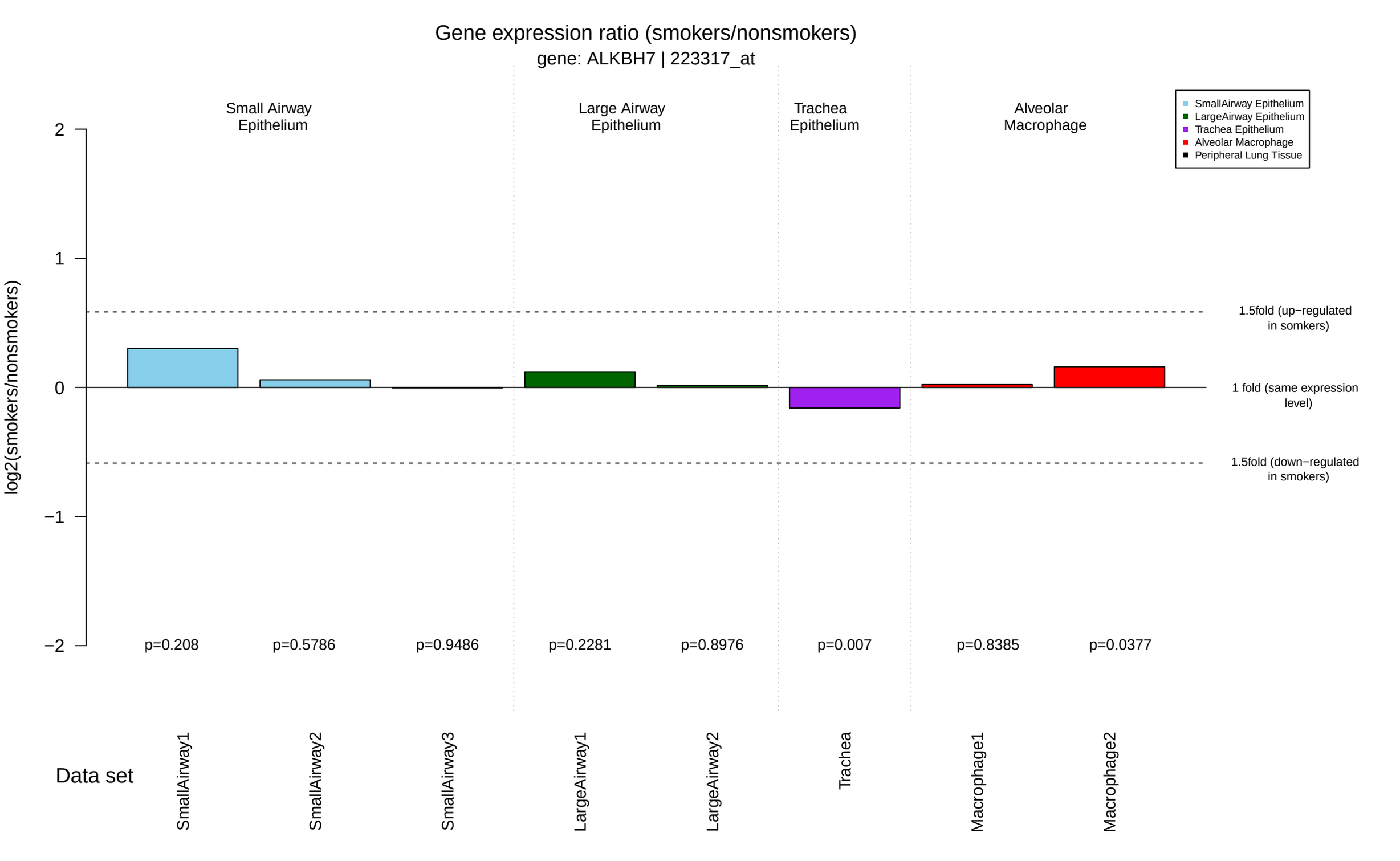
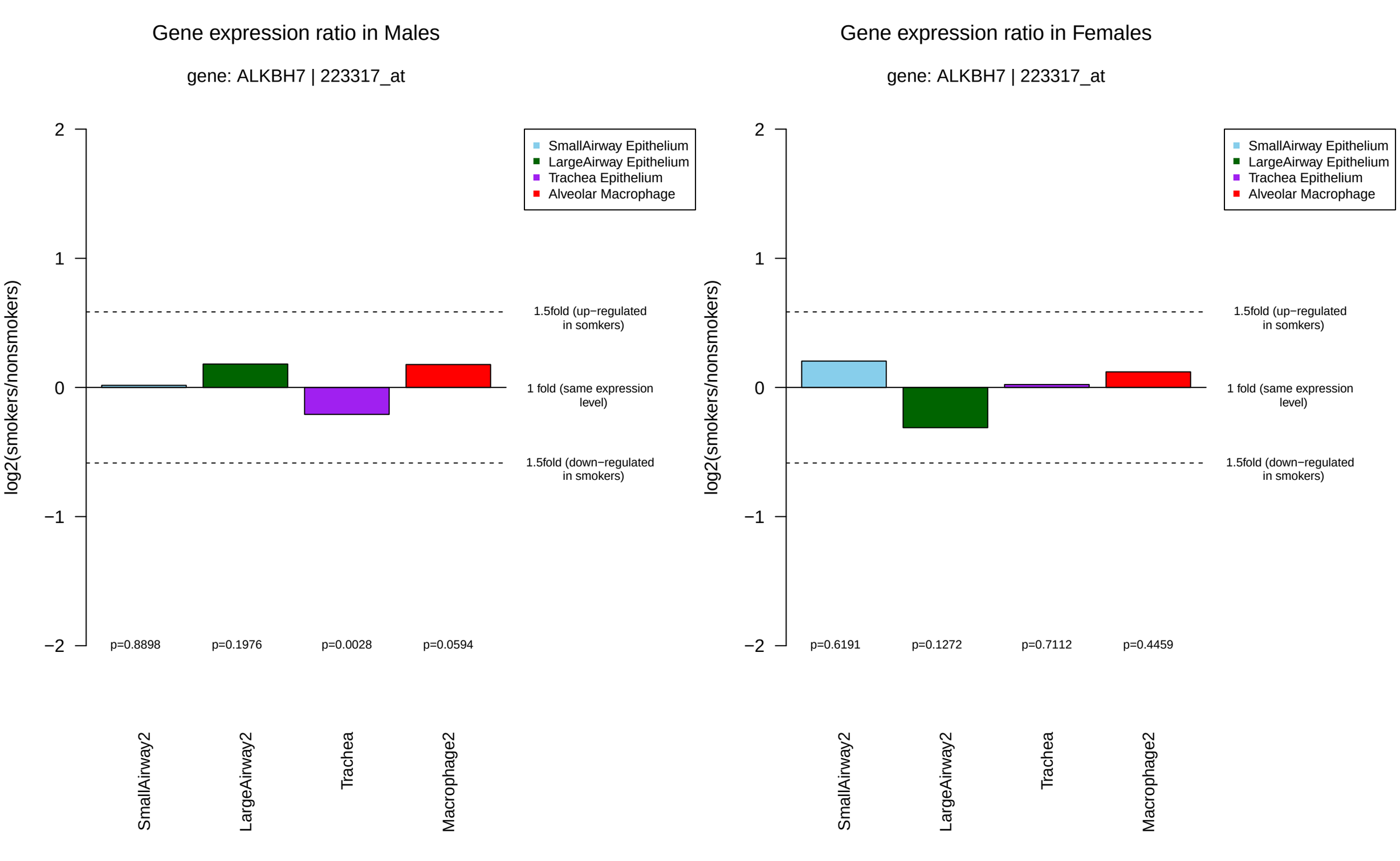
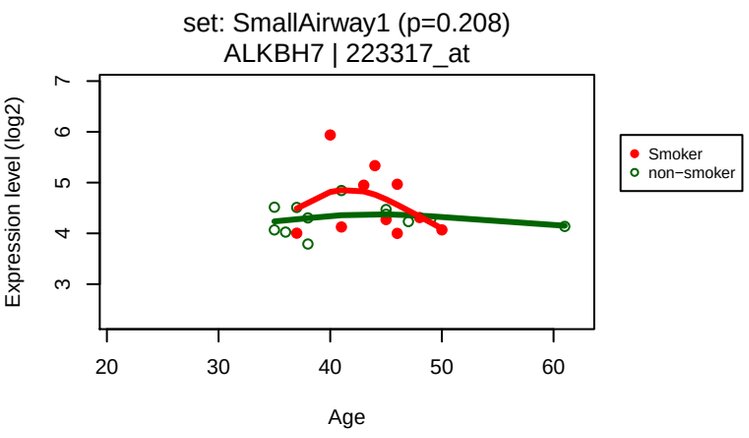
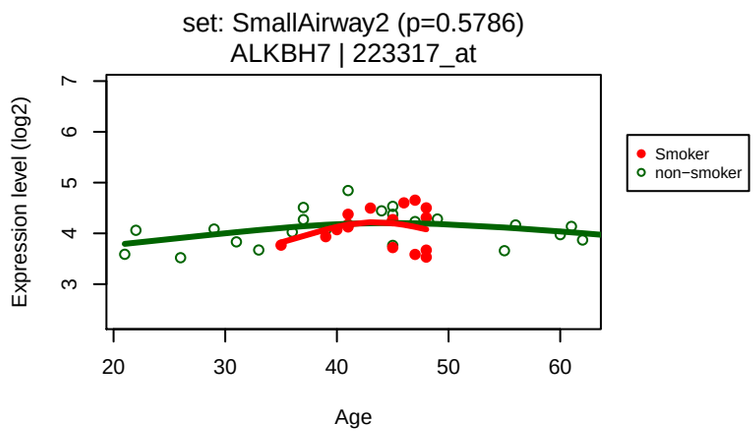
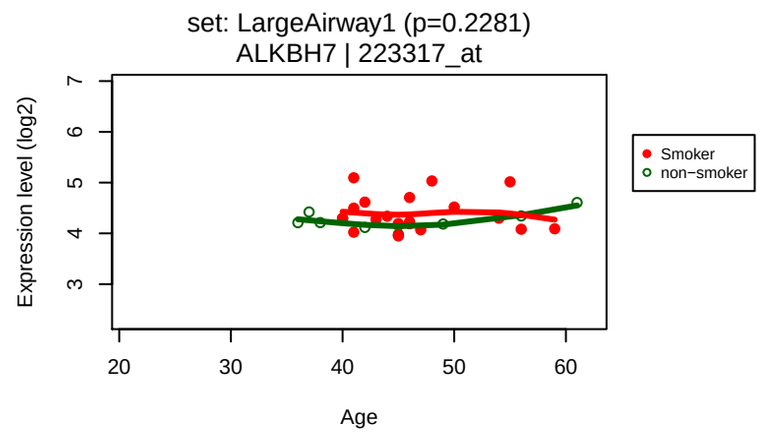

Gene expression of smokers and nonsmokers
gene: ALKBH7 | 223317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.208
p=0.5786
p=0.9486
p=0.2281
p=0.8976
p=0.007
p=0.8385
p=0.0377
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

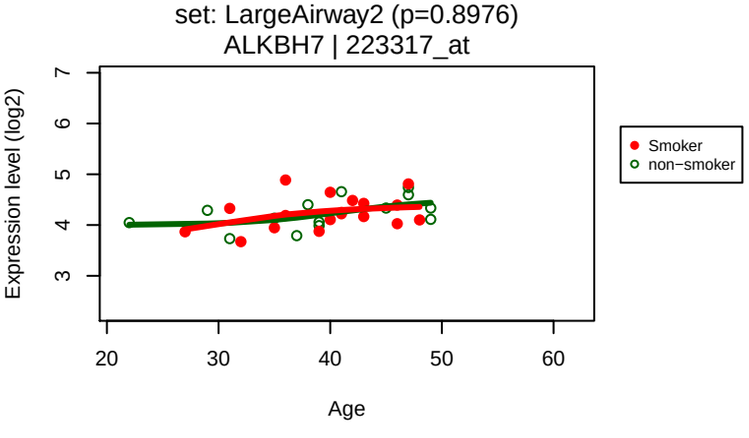

Gene expression ratio (smokers/nonsmokers)
gene: ALKBH7 | 223317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.208
p=0.5786
p=0.9486
p=0.2281
p=0.8976
p=0.007
p=0.8385
p=0.0377
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

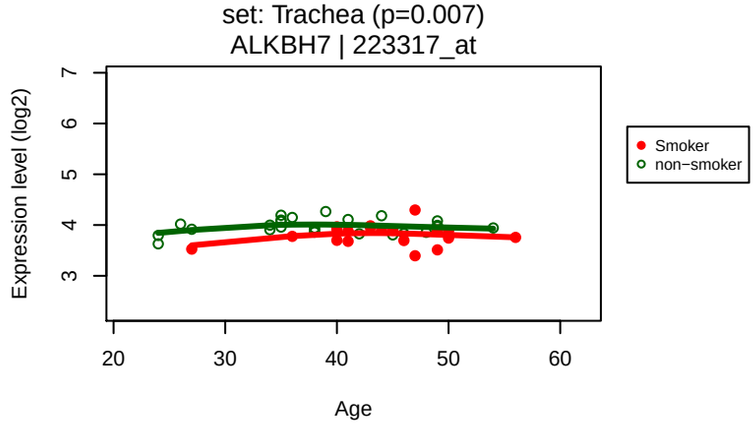

Gene expression ratio in Males
gene: ALKBH7 | 223317_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8898
p=0.1976
p=0.0028
p=0.0594
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ALKBH7 | 223317_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6191
p=0.1272
p=0.7112
p=0.4459
SmallAirway2
LargeAirway2
Trachea
Macrophage2

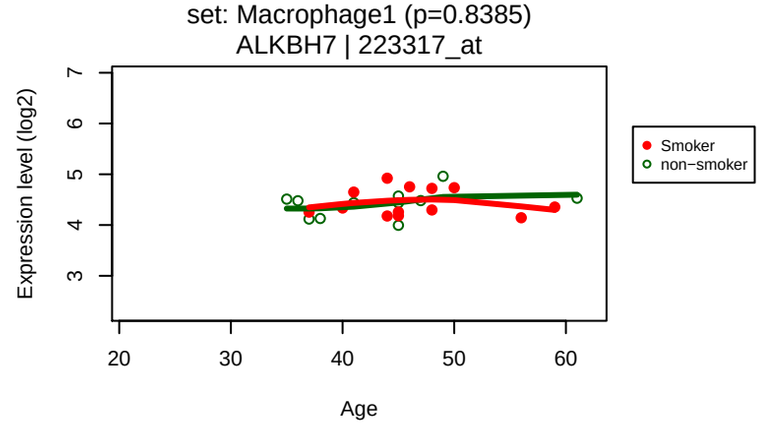

set: SmallAirway1 (p=0.208)
ALKBH7 | 223317_at
Expression level (log2)
Age
Smoker
non−smoker

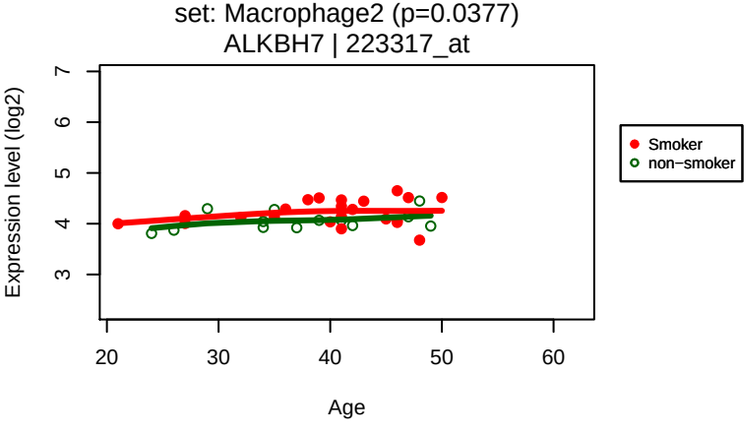

set: SmallAirway2 (p=0.5786)
ALKBH7 | 223317_at
Expression level (log2)
Age
Smoker
non−smoker

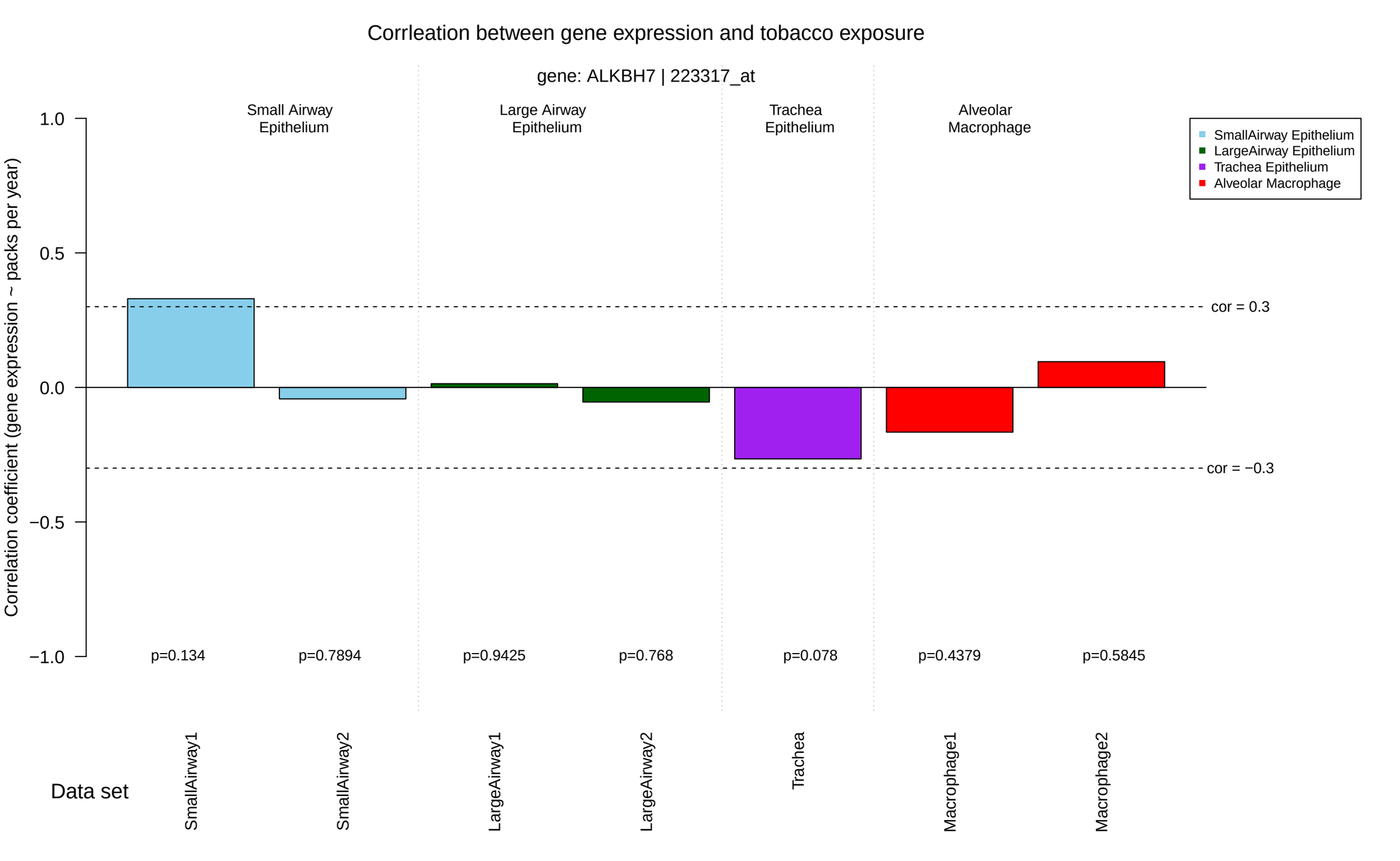

set: LargeAirway1 (p=0.2281)
ALKBH7 | 223317_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8976)
ALKBH7 | 223317_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.007)
ALKBH7 | 223317_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8385)
ALKBH7 | 223317_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0377)
ALKBH7 | 223317_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ALKBH7 | 223317_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.134
p=0.7894
p=0.9425
p=0.768
p=0.078
p=0.4379
p=0.5845
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2